

Engine Cranks But Will Not Start

Symptoms

- The engine will crank but **not** start when start button is pressed.

How To Use This Tree

This symptom tree can be used to troubleshoot display power symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform depending on the symptom.

Shoptalk

- To initiate engine crank, the following panel parameters **must** be met:
- The system enable switch is turned to the ON position
 - The engine is stopped
 - Main throttle and backup throttle are in neutral position
 - Battery disconnect switch is turned ON.
- Possible causes are:
- Stop is shorted or open
 - Fire suppression circuit open
 - Starter lockout on engine is engaged.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.).	
	STEP 1A. Check the starter lockout relay in the C.I.B., if equipped.	LED for starter lockout illuminated?

STEPS		SPECIFICATIONS
	STEP 1B. Check the original equipment manufacturer (OEM) interface wiring harness for an open circuit in the fire suppression circuit.	Less than 10 ohms?
	STEP 1C. Check for engine control module (ECM) fault codes.	Active fault codes on ECM?

STEP 1. Check the C.I.B.

STEP 1A. Check the starter lockout relay in the C.I.B. (if equipped).

Conditions :

- Open up the C.I.B. Refer to Procedure 015-023 in Section 15 (</qs3/pubsys2/xml/en/procedures/513/513-015-023.html>).

Action	Specification/Repair	Next Step
Check the starter lockout relay in the C.I.B. • Locate the starter lockout relay LED. Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-023.html)	LED for starter lockout illuminated? YES Repair : The ECM has locked out the engine from starting. Investigate engine with INSITE™ electronic service tool for related fault codes.	Repair complete.
	LED for starter lockout illuminated? NO	1B

STEP 1B. Check the OEM Interface wiring harness for an open circuit in the fire suppression circuit.

Conditions :

- Turn system enable switch OFF.
- Disconnect OEM interface wiring harness from the C.I.B.

Action	Specification/Repair	Next Step
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Check the OEM Interface wiring harness for an open circuit in the fire suppression circuit. Measure the resistance across the fire suppression circuit pins 11 and 12 on the OEM Interface wiring harness connector (mating to the C.I.B). Refer to the circuit diagram or wiring diagram for connector pin identification. Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Less than 10 ohms? YES	2A
	Less than 10 ohms? NO Repair : An open in the fire suppression circuit has been detected. Repair the OEM interface wiring harness and interconnects. Refer to Procedure 015-104 in Section 15. (/qs3/pubsys2/xml/en/procedures/513/513-015-104.html)	Repair complete.

STEP 1C. Check for ECM fault codes.

Conditions : - Turn keyswitch ON. - Connect INSITE™ electronic service tool.		
Action	Specification/Repair	Next Step
Check for active fault codes. Use INSITE™ electronic service tool to read the fault codes.	Active fault codes on ECM? YES	Go to the appropriate fault code troubleshooting tree.
	Active fault codes on ECM? NO	Refer to Engine Difficult To Start Or Will Not Start troubleshooting symptom tree in Section TS in the appropriate service manual.

